

# PRANJAL GUPTA

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## Skills

- Programming Languages: Python, Java, JavaScript
- Data Structures, Algorithms & Object-Oriented Programming
- Full Stack Web Development: React, Node.js, Express.js, Flask, REST APIs
- Databases: MySQL, MongoDB
- Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, Keras, OpenCV
- Computer Vision & Generative AI: CNN, LangChain, RAG Basics
- Cloud & DevOps: AWS (EC2, S3 basics), Docker, Git, GitHub, Linux, CLI
- Developer Tools: Postman, VS Code, Jupyter Notebook, OpenProject, Taiga

## Languages

**Hindi:** Native  
**English:** Proficient (C2)  
**Japanese:** Beginner

## Certifications

- CIN:PIT/JUL25/03934 - Full Stack Development
- Indian DefSpace Symposium (Apr 2025)

## Summary

Undergraduate Computer Science student specializing in Artificial Intelligence and Machine Learning with experience in machine learning, neural networks, computer vision, and full stack development. Built projects involving sign language interpretation, predictive modeling, REST APIs, and real-time web applications using modern development frameworks.

Actively exploring Generative AI, LangChain, RAG pipelines, AI agents, and workflow automation, with a strong interest in building practical AI-driven systems and intelligent applications.

## Experience

**Full Stack Web Developer Intern** *Prodigy Infotech* July 2025

- Developed multiple full-stack web applications using React.js, Node.js, Express.js, and MongoDB.
- Built features including authentication systems, REST API integration, and real-time chat functionality.
- Debugged and optimized application logic, improving functionality and user experience.

**Volunteer** *Indian DefSpace Symposium 2025*

- Assisted in coordination and logistics for a national-level space science symposium.
- Interacted with researchers, speakers, and industry professionals during technical sessions and events.

## Education and Training

**B.Tech in Computer Science Engineering (AI & ML Specialization)**  
2023-2027

Faculty of Technology – North Campus, University of Delhi, India  
*First Year GPA: 7.0 / 10.0 | Second Year GPA: 7.8 / 10.0*

**Class XII (CBSE)** *2021-2022*

St. Andrews Public School  
*Percentage: 91%*

**Class X (CBSE)** *2019-2020*

St. Andrews Public School  
*Percentage: 93%*

## Projects

- **Hybrid Sequential Feature Fusion Pipeline for Audio Classification** – Engineered an end-to-end classification pipeline differentiating biological and synthetic speech by fusing 1536-D Wav2Vec2 latent embeddings with deterministic acoustic metrics (MFCC, YIN, Spectral Rolloff).
- **Real-Time Communication Chat Application** – Built a full-stack real-time chat system using React, Node.js, Socket.IO, MongoDB, and JWT authentication.
- **Space Debris Position Prediction (SGP4 + LSTM)** – Compared analytical SGP4 orbital prediction with LSTM-based sequence modeling; LSTM did not outperform SGP4, highlighting data and generalization limits.
- **Indian Sign Language Translation System** – Built a bidirectional Sign-to-Speech and Speech-to-Sign pipeline using computer vision and NLP for accessibility-focused applications.